Saving model on Tensorflow 2.7.0 with data augmentation layer

Asked 2 years, 2 months ago Modified 2 years ago Viewed 4k times

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I am getting an error when trying to save a model with data augmentation layers with Tensorflow version 2.7.0.

Here is the code of data augmentation:

```
input_shape_rgb = (img_height, img_width, 3)
data_augmentation_rgb = tf.keras.Sequential(
  [
    layers.RandomFlip("horizontal"),
    layers.RandomFlip("vertical"),
    layers.RandomRotation(0.5),
    layers.RandomZoom(0.5),
    layers.RandomContrast(0.5),
    RandomColorDistortion(name='random_contrast_brightness/none'),
  ]
)
```

Now I build my model like this:

```
# Build the model
input_shape = (img_height, img_width, 3)

model = Sequential([
  layers.Input(input_shape),
  data_augmentation_rgb,
  layers.Rescaling((1./255)),

  layers.Conv2D(16, kernel_size, padding=padding, activation='relu', strides=1,
     data_format='channels_last'),
  layers.MaxPooling2D(),
  layers.BatchNormalization(),

  layers.Conv2D(32, kernel_size, padding=padding, activation='relu'), # best 4
  layers.MaxPooling2D(),
  layers.BatchNormalization(),

  layers.Conv2D(64, kernel_size, padding=padding, activation='relu'), # best 3
  layers.MaxPooling2D(),
  layers.BatchNormalization(),

  layers.Conv2D(128, kernel_size, padding=padding, activation='relu'), # best 3
  layers.MaxPooling2D(),
  layers.BatchNormalization(),

  layers.Flatten(),
  layers.Dense(128, activation='relu'), # best 1
  layers.Dropout(0.1),
  layers.Dense(128, activation='relu'), # best 1
```

```
 layers.Dropout(0.1),
 layers.Dense(64, activation='relu'), # best 1
 layers.Dropout(0.1),
 layers.Dense(num_classes, activation = 'softmax')
])

model.compile(loss='categorical_crossentropy', optimizer='adam',metrics=metrics)
model.summary()
```

Then after the training is done I just make:

```
model.save("./")
```

And I'm getting this error:

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
<ipython-input-84-87d3f09f8bee> in <module>()
----> 1 model.save("./")


/usr/local/lib/python3.7/dist-packages/keras/utils/traceback_utils.py in
 error_handler(*args, **kwargs)
 65     except Exception as e:  # pylint: disable=broad-except
 66       filtered_tb = _process_traceback_frames(e.__traceback__)
 ---> 67       raise e.with_traceback(filtered_tb) from None
 68     finally:
 69       del filtered_tb

 /usr/local/lib/python3.7/dist-
 packages/tensorflow/python/saved_model/function_serialization.py in
 serialize_concrete_function(concrete_function, node_ids, coder)
 66   except KeyError:
 67     raise KeyError(
 ---> 68         f"Failed to add concrete function '{concrete_function.name}' to
 object-"
 69         f"based SavedModel as it captures tensor {capture!r} which is
 unsupported"
 70         " or not reachable from root. "

 KeyError: "Failed to add concrete function
 'b'__inference_sequential_46_layer_call_fn_662953'' to object-based SavedModel as it
 captures tensor <tf.Tensor: shape=(), dtype=resource, value=<Resource Tensor>> which
 is unsupported or not reachable from root. One reason could be that a stateful
 object or a variable that the function depends on is not assigned to an attribute of
 the serialized trackable object (see SaveTest.test_captures_unreachable_variable)."
```

I inspected the reason of getting this error by changing the architecture of my model and I just found that reason came from the data_augmentation layer since the `RandomFlip` and `RandomRotation` and others are changed from `layers.experimental.prepocessing.RandomFlip` to `layers.RandomFlip` , but still the error appears.

python tensorflow keras deep-learning data-augmentation

2 Answers

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This seems to be a bug in Tensorflow 2.7 when using `model.save` combined with the parameter `save_format="tf"`, which is set by default. The layers `RandomFlip`, `RandomRotation`, `RandomZoom`, and `RandomContrast` are causing the problems, since they are not serializable. Interestingly, the `Rescaling` layer can be saved without any problems. A workaround would be to simply save your model with the older Keras H5 format `model.save("test", save_format='h5')`:

```
import tensorflow as tf
import numpy as np

class RandomColorDistortion(tf.keras.layers.Layer):
    def __init__(self, contrast_range=[0.5, 1.5],
                 brightness_delta=[-0.2, 0.2], **kwargs):
        super(RandomColorDistortion, self).__init__(**kwargs)
        self.contrast_range = contrast_range
        self.brightness_delta = brightness_delta

    def call(self, images, training=None):
        if not training:
            return images
        contrast = np.random.uniform(
            self.contrast_range[0], self.contrast_range[1])
        brightness = np.random.uniform(
            self.brightness_delta[0], self.brightness_delta[1])

        images = tf.image.adjust_contrast(images, contrast)
        images = tf.image.adjust_brightness(images, brightness)
        images = tf.clip_by_value(images, 0, 1)
        return images

    def get_config(self):
        config = super(RandomColorDistortion, self).get_config()
        config.update({"contrast_range": self.contrast_range, "brightness_delta":
self.brightness_delta})
        return config

input_shape_rgb = (256, 256, 3)
data_augmentation_rgb = tf.keras.Sequential(
  [
    tf.keras.layers.RandomFlip("horizontal"),
    tf.keras.layers.RandomFlip("vertical"),
    tf.keras.layers.RandomRotation(0.5),
    tf.keras.layers.RandomZoom(0.5),
    tf.keras.layers.RandomContrast(0.5),
    RandomColorDistortion(name='random_contrast_brightness/none'),
  ]
)
input_shape = (256, 256, 3)
padding = 'same'
```

```
kernel_size = 3
model = tf.keras.Sequential([
  tf.keras.layers.Input(input_shape),
  data_augmentation_rgb,
  tf.keras.layers.Rescaling((1./255)),
  tf.keras.layers.Conv2D(16, kernel_size, padding=padding, activation='relu',
strides=1,
      data_format='channels_last'),
  tf.keras.layers.MaxPooling2D(),
  tf.keras.layers.BatchNormalization(),

  tf.keras.layers.Conv2D(32, kernel_size, padding=padding, activation='relu'), # best 4
  tf.keras.layers.MaxPooling2D(),
  tf.keras.layers.BatchNormalization(),

  tf.keras.layers.Conv2D(64, kernel_size, padding=padding, activation='relu'), # best 3
  tf.keras.layers.MaxPooling2D(),
  tf.keras.layers.BatchNormalization(),

  tf.keras.layers.Conv2D(128, kernel_size, padding=padding, activation='relu'), # best
3
  tf.keras.layers.MaxPooling2D(),
  tf.keras.layers.BatchNormalization(),

  tf.keras.layers.Flatten(),
  tf.keras.layers.Dense(128, activation='relu'), # best 1
  tf.keras.layers.Dropout(0.1),
  tf.keras.layers.Dense(128, activation='relu'), # best 1
  tf.keras.layers.Dropout(0.1),
  tf.keras.layers.Dense(64, activation='relu'), # best 1
  tf.keras.layers.Dropout(0.1),
  tf.keras.layers.Dense(5, activation = 'softmax')
 ])

model.compile(loss='categorical_crossentropy', optimizer='adam')
model.summary()
model.save("test", save_format='h5')
```

Loading your model with your custom layer would look like this then:

```
model = tf.keras.models.load_model('test.h5', custom_objects={'RandomColorDistortion':
RandomColorDistortion})
```

where `RandomColorDistortion` is the name of your custom layer.

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edited Feb 4, 2022 at 17:25

answered Nov 15, 2021 at 13:53



AloneTogether
26.3k 5 20 39

1 when trying to use h5 format i get another error of type NotImpleentedError. – moumed Nov 24, 2021 at 17:15

1 Nope the error you are getting has nothing to do with your original issue. As already mentioned in GitHub, you have to add a config to your custom layer if you plan to save it later. Check this post for example: stackoverflow.com/questions/62280161/... or just look at the config of the custom layer in my answer. – AloneTogether Nov 24, 2021 at 17:51

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ah yep of course, I forgot this detai ng the model save to load later the model and convert it to .tflite, wher ut the customer layer, I will make the error on github issue and we wi eTogether – moumed Nov 24, 2021 at 19:35

1 Updated answer, did it work? – AloneTogether Nov 25, 2021 at 14:18

1 When doing so, I have CustomMaskWarning, I seems to work but I hope it doesn't break the model – SashimiDélicieux Jan 19, 2022 at 11:10

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You can also downgrade Keras and Tensorflow to version 2.6.

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answered Nov 24, 2021 at 16:29



Alx-net

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1 this is a solution that i'am using actually, but I want to use the last TF version – moumed Nov 24, 2021 at 17:12